

Conformal Prediction with Missing Values

Margaux Zaffran

Conformal Inference Reading Group – IRT Saint-Exupéry

October 24, 2023



Who am I?

- 3rd (last) year statistics PhD Student, @ INRIA & École Polytechnique (Paris)
- Funded by Électricité de France (*French main electricity producer and supplier*)
- My advisors:



**Aymeric
Dieuleveut**
École Polytechnique



Olivier Féron
EDF R&D
FiME



Yannig Goude
EDF R&D
LMO



Julie Josse
PreMeDICaL
INRIA

- Research interests:
 - Distribution-free uncertainty quantification
 - Time series data
 - Missing values
 - Societal applications (energy, environmental and medical domains)

Conformal Prediction with Missing Covariates



Aymeric Dieuleveut

École

Polytechnique

Paris - France



Julie Josse

PreMeDICaL

INRIA

Montpellier - France



Yaniv Romano

Technion - Israel Institute
of Technology

Haifa - Israel

Introduction to missing values

Quantifying predictive uncertainty with missing values

Conclusion

- 30 hospitals
- More than 30 000 trauma patients
- 4 000 new patients per year
- 250 continuous and categorical variables
 - ↪ Many useful statistical tasks

Predict the level of blood platelets upon arrival at hospital, given 7 pre-hospital features.

These covariates are not always observed.

Missing values: ubiquitous in data science practice

Data: $(X^{(k)}, Y^{(k)})_{k=1}^n \in (\mathbb{R}^d \times \mathbb{R})^n$

Y	X_1	X_2	X_3	X_4	X_5	X_6
8.26	0.72	0.18	0.55	0.05	0.73	0.50
19.41	0.60	0.58	NA	NA	NA	0.40
19.75	0.54	0.43	0.96	0.77	0.06	0.66
7.32	NA	0.19	NA	0.02	0.83	0.04
13.55	0.65	0.69	0.50	0.15	NA	0.87
20.75	0.43	0.74	0.61	0.72	0.52	0.35
9.26	0.89	NA	0.84	0.01	0.73	NA
9.68	0.963	0.45	0.65	0.04	0.06	NA

If each entry has a probability 0.01 of being missing:

$d = 6 \rightarrow \approx 94\%$ of rows kept

$d = 300 \rightarrow \approx 5\%$ of rows kept

*One of the ironies of Big Data is that missing data play an ever more significant role.*¹

¹Zhu et al. (2019), *High-dimensional PCA with heterogeneous missingness*, JRSS B

Handling missing values depends on pattern and mechanism

- $(X, Y) \in \mathbb{R}^d \times \mathbb{R}$ random variables.
- $M \in \{0, 1\}^d$ is defined as $M_j = 1 \Leftrightarrow X_j$ is missing.
 M is called the **mask** or the **missing pattern**.

Example

We observe $(-1, \text{NA}, \text{NA})$. Then $m = (0, 1, 1)$ and $X_{\text{obs}(m)} = (-1)$.

There are 2^d **patterns** (statistical and computational challenges).

- Three **mechanisms**² can generate missing values.
 - $\hookrightarrow$ **Missing Completely At Random** (MCAR): $\mathbb{P}(M = m|X) = \mathbb{P}(M = m)$
for all $m \in \{0, 1\}^d$. $M \perp\!\!\!\perp X$, missingness does not depend on the variables.

²Rubin (1976), *Inference and missing data*, Biometrika

Supervised learning with missing values: impute-then-regress

Impute-then-regress procedures are widely used.

1. Replace NA using an **imputation function** ϕ (e.g. the mean).
2. Train your algorithm (Random Forest, Neural Nets, etc.) on the **imputed**

$$\text{data: } \left\{ \underbrace{\phi\left(X^{(k)}, M^{(k)}\right)}_{\text{imputed } X^{(k)}}, Y^{(k)} \right\}_{k=1}^n.$$

↪ we consider an **impute-then-regress** pipeline in this work.

- ✓ Le Morvan et al. (2021)³ show that for any deterministic imputation and universal learner this procedure is Bayes-consistent.
- ✗ Ayme et al. (2022)⁴ show that even for very simple distributions (linear model, Gaussian noise), this rate of convergence may suffer from curse of dimensionality.

³ Le Morvan, Josse, Scornet & Varoquaux (2021), *What's a good imputation to predict with missing values?*, NeurIPS

⁴ Ayme, Boyer, Dieuleveut & Scornet (2022), *Near-optimal rate of consistency for linear models with missing values*, ICML

Introduction to missing values

Quantifying predictive uncertainty with missing values

Conformalized Quantile Regression

Impute-then-Regress+Conformalization

Missing Data Augmentation

Experimental results

Conclusion

Predictive uncertainty quantification with missing values

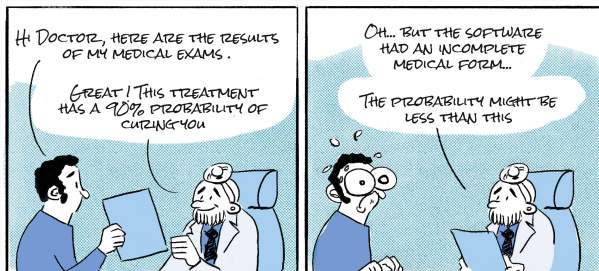
Goal: predict $Y^{(n+1)}$ with **confidence** $1 - \alpha$, i.e. build the smallest $\mathcal{C}_\alpha$ such that:

1. Marginal Validity (MV)

$$\mathbb{P} \left\{ Y^{(n+1)} \in \mathcal{C}_\alpha \left(X^{(n+1)}, M^{(n+1)} \right) \right\} \geq 1 - \alpha. \quad (\text{MV})$$

2. Mask-Conditional-Validity (MCV)

$$\forall m \in \{0, 1\}^d : \mathbb{P} \left\{ Y^{(n+1)} \in \mathcal{C}_\alpha \left(X^{(n+1)}, m \right) \mid M^{(n+1)} = m \right\} \geq 1 - \alpha. \quad (\text{MCV})$$



Illustrations @theo.reminger

Introduction to missing values

Quantifying predictive uncertainty with missing values

Conformalized Quantile Regression

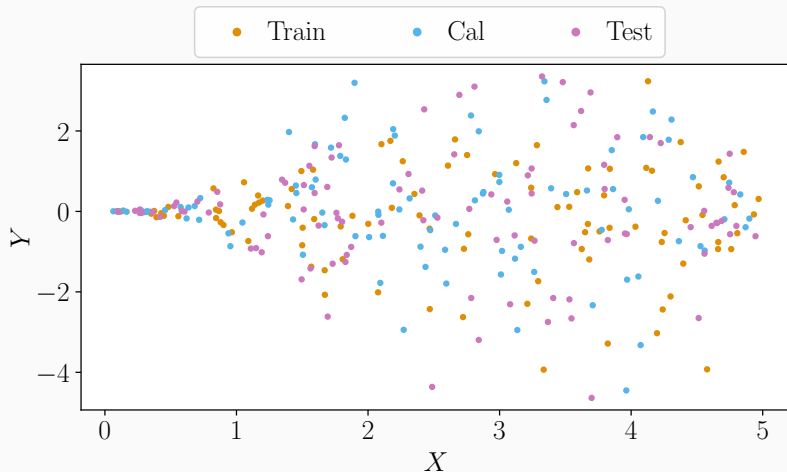
Impute-then-Regress+Conformalization

Missing Data Augmentation

Experimental results

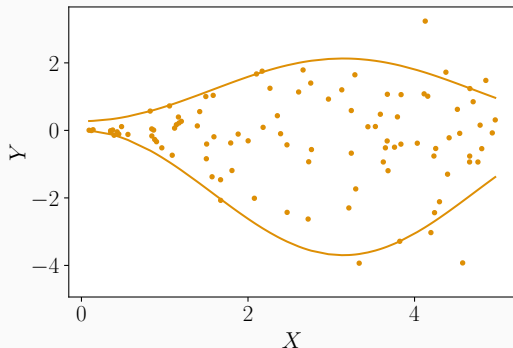
Conclusion

Conformalized Quantile Regression (CQR)⁴: toy example



⁴Romano et al. (2019), *Conformalized Quantile Regression*, NeurIPS

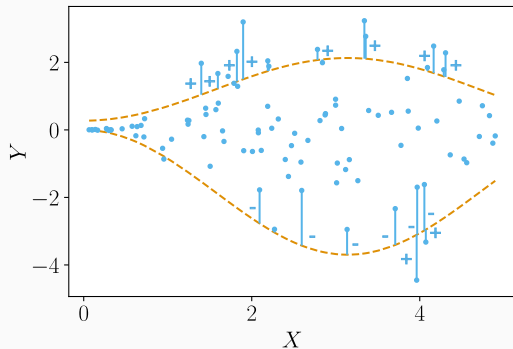
Conformalized Quantile Regression (CQR)⁴: training step



► Learn (or get) $\widehat{QR}_{\text{lower}}$ and $\widehat{QR}_{\text{upper}}$

⁴Romano et al. (2019), *Conformalized Quantile Regression*, NeurIPS

Conformalized Quantile Regression (CQR)⁴: calibration step

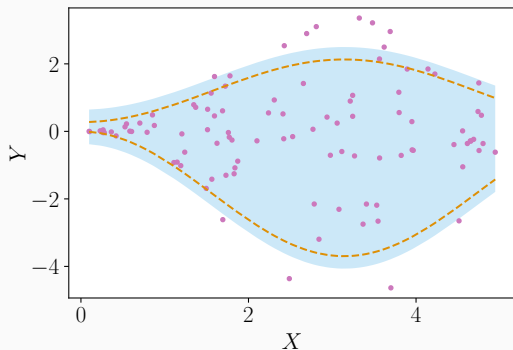


- ▶ Predict with $\widehat{QR}_{lower}$ and $\widehat{QR}_{upper}$
- ▶ Get the scores $\mathcal{S} = \{S^{(k)}\}_{\text{Cal}} \cup \{+\infty\}$
- ▶ Compute the $(1 - \alpha)$ empirical quantile of $\mathcal{S}$, noted $q_{1-\alpha}(\mathcal{S})$

$$\hookrightarrow S^{(k)} := \max \left\{ \widehat{QR}_{lower} \left(X^{(k)} \right) - Y^{(k)}, Y^{(k)} - \widehat{QR}_{upper} \left(X^{(k)} \right) \right\}$$

⁴Romano et al. (2019), *Conformalized Quantile Regression*, NeurIPS

Conformalized Quantile Regression (CQR)⁴: prediction step



► Predict with $\widehat{QR}_{\text{lower}}$ and $\widehat{QR}_{\text{upper}}$

► Build

$$\widehat{C}_\alpha(x) = [\widehat{QR}_{\text{lower}}(x) - q_{1-\alpha}(S); \widehat{QR}_{\text{upper}}(x) + q_{1-\alpha}(S)]$$

⁴Romano et al. (2019), *Conformalized Quantile Regression*, NeurIPS

CQR: theoretical guarantees

CQR enjoys finite sample guarantees proved in Romano et al. (2019), as a particular case of Conformal Prediction (CP).

Theorem

Suppose $(X^{(k)}, Y^{(k)})_{k=1}^{n+1}$ are *exchangeable (or i.i.d.)*. CQR applied on $(X^{(k)}, Y^{(k)})_{k=1}^n$ outputs $\hat{C}_\alpha(\cdot)$ such that:

$$\mathbb{P} \left\{ Y^{(n+1)} \in \hat{C}_\alpha \left(X^{(n+1)} \right) \right\} \geq 1 - \alpha.$$

Additionally, if the scores $\{S^{(k)}\}_{k \in \text{Cal}}$ are a.s. distinct:

$$\mathbb{P} \left\{ Y^{(n+1)} \in \hat{C}_\alpha \left(X^{(n+1)} \right) \right\} \leq 1 - \alpha + \frac{1}{\#\text{Cal} + 1}.$$

✓ Distribution-free, only requires exchangeability

✓ Any quantile regression algorithm (neural nets, random forest...)

✓ Finite sample

✗ Marginal coverage: $\mathbb{P} \left\{ Y^{(n+1)} \in \hat{C}_\alpha \left(X^{(n+1)} \right) \mid \cancel{X^{(n+1)} = x} \right\} \geq 1 - \alpha$

Introduction to missing values

Quantifying predictive uncertainty with missing values

Conformalized Quantile Regression

Impute-then-Regress+Conformalization

Missing Data Augmentation

Experimental results

Conclusion

CP is marginally valid (MV) after imputation

To apply conformal prediction we need **exchangeable** data.

Lemma (Zaffran et al. (2023a))

Assume $(X^{(k)}, M^{(k)}, Y^{(k)})_{k=1}^n$ are i.i.d. (or exchangeable).

Then, for *any missing mechanism*, for almost *all imputation function*⁵ ϕ :

$(\phi(X^{(k)}, M^{(k)}), Y^{(k)})_{k=1}^n$ are **exchangeable**.

$\Rightarrow$ CQR, and Conformal Prediction, applied on an imputed data set still enjoys marginal guarantees⁶:

$$\mathbb{P} \left\{ Y^{(n+1)} \in \hat{C}_\alpha \left(X^{(n+1)}, M^{(n+1)} \right) \right\} \geq 1 - \alpha.$$

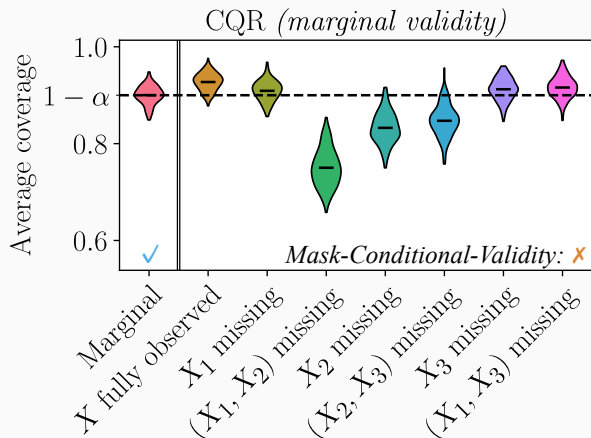
⁵Even if the imputation is not accurate, the guarantee will hold.

⁶The upper bound also holds under continuously distributed scores.

CQR is marginally valid on imputed data sets

$$Y = \beta^T X + \varepsilon,$$

$\beta = (1, 2, -1)^T$, $\varepsilon \perp\!\!\!\perp X$, X and ε Gaussian, 20% uniform MCAR missing values.



Warning: the predictive intervals cover properly **marginally**, but suffer from high **disparities depending on the missing patterns**.

Missing values induce heteroskedasticity

Gaussian linear model

- $Y = \beta^T X + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2) \perp (X, M)$, $\beta \in \mathbb{R}^d$.
- for all $m \in \{0, 1\}^d$, there exist μ^m and Σ^m such that $X|(M = m) \sim \mathcal{N}(\mu^m, \Sigma^m)$.

↪ **oracle** intervals: smallest predictive interval when the distribution of $Y|(X, M)$ is known

Proposition (Oracle int. under Gaussian lin. mod., Zaffran et al. (2023a))

$$\mathcal{L}_\alpha^*(m) = 2 \times q_{1-\alpha/2}^{\mathcal{N}(0,1)} \times \sqrt{\beta_{\text{mis}(m)}^T \Sigma_{\text{mis}|\text{obs}}^m \beta_{\text{mis}(m)} + \sigma_\varepsilon^2}.$$

- Even with an homoskedastic noise, missingness generates **heteroskedasticity**
- The uncertainty increases when **missing values are associated with larger regression coefficients** (i.e. the most predictive variables)

Goals reminder: achieve MCV!

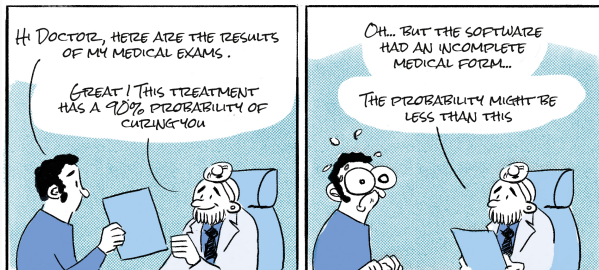
Goal: predict $Y^{(n+1)}$ with **confidence** $1 - \alpha$, i.e. build the smallest $\mathcal{C}_\alpha$ such that:

1. Marginal Validity (MV) ✓

$$\mathbb{P} \left\{ Y^{(n+1)} \in \mathcal{C}_\alpha \left(X^{(n+1)}, M^{(n+1)} \right) \right\} \geq 1 - \alpha. \quad (\text{MV})$$

2. Mask-Conditional-Validity (MCV) ✗

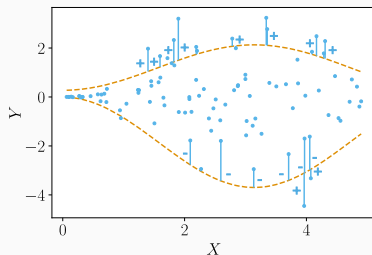
$$\forall m \in \{0, 1\}^d : \mathbb{P} \left\{ Y^{(n+1)} \in \mathcal{C}_\alpha \left(X^{(n+1)}, m \right) \mid M^{(n+1)} = m \right\} \geq 1 - \alpha. \quad (\text{MCV})$$



Illustrations @theo.reminger

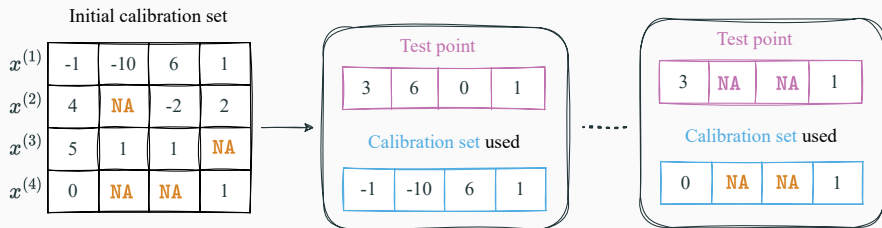
Conformalization step is independent of the important variable: the mask!

Observation: the α -correction term is computed among all the data points, regardless of their mask!



Warning: 2^d possible masks

⇒ Splitting the calibration set by mask (*Mondrian type*) is infeasible (lack of data)!



Introduction to missing values

Quantifying predictive uncertainty with missing values

Conformalized Quantile Regression

Impute-then-Regress+Conformalization

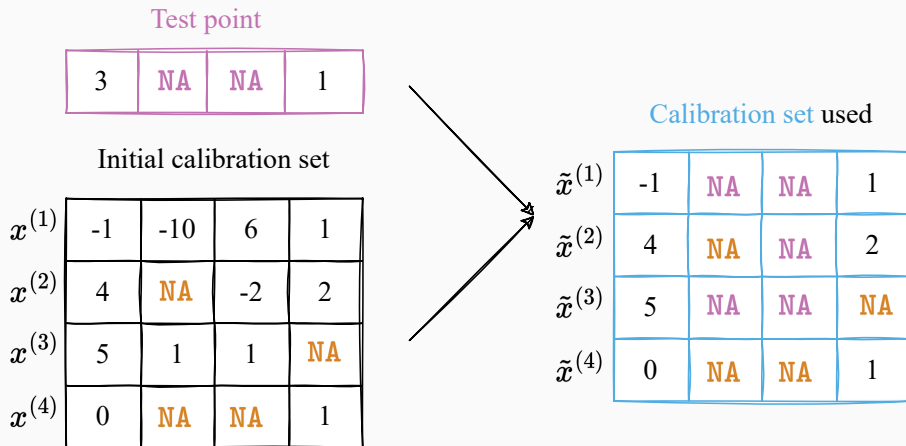
Missing Data Augmentation

Experimental results

Conclusion

Missing Data Augmentation (MDA) of the calibration set

Idea: for each **test point**, modify the **calibration points** to mimic the **test mask**



Algorithms: MDA with **Exact** masking or with **Nested** masking.

Introduction to missing values

Quantifying predictive uncertainty with missing values

Conformalized Quantile Regression

Impute-then-Regress+Conformalization

Missing Data Augmentation

MDA with Exact masking

MDA with Nested masking

Experimental results

Conclusion

CP-MDA with Exact masking

Test point

3	NA	NA	1
---	----	----	---

Initial calibration set

$x^{(1)}$	-1	-10	6	1
$x^{(2)}$	4	NA	-2	2
$x^{(3)}$	5	1	1	NA
$x^{(4)}$	0	NA	NA	1



Calibration set used

$\tilde{x}^{(1)}$	-1	NA	NA	1
$\tilde{x}^{(2)}$	4	NA	NA	2
$\tilde{x}^{(3)}$	[Hatched area]			
$\tilde{x}^{(4)}$	0	NA	NA	1

$\# \text{Cal}^{\text{M}(\text{test})}$ observations

CQR-MDA with exact masking in words

1. Split the training set into a **proper training set** and **calibration set**
2. Train the imputation function on the **proper training set**
3. Impute the **proper training set**
4. Train the **quantile regressors** on the imputed **proper training set**
5. For a test point $(X^{(n+1)}, M^{(n+1)})$:

5.1 For each $j \in \llbracket 1, d \rrbracket$ s.t. $M_j^{(n+1)} = 1$, set $\tilde{M}_j^{(k)} = 1$
for k in **Cal** s.t. $M^{(k)} \subset M^{(n+1)}$

5.2 Impute the new **calibration set**

5.3 Compute the **calibration correction**, i.e. $q_{1-\alpha}(\mathcal{S})$

5.4 Impute the **test point**

5.5 Predict with the **quantile regressors** and the **correction** previously obtained,
 $q_{1-\alpha}(\mathcal{S})$

	3	NA	NA	1
$\tilde{x}^{(1)}$	-1	NA	NA	1
$\tilde{x}^{(2)}$	4	NA	NA	2
$\tilde{x}^{(3)}$				
$\tilde{x}^{(4)}$	0	NA	NA	1

Theorem (CP-MDA-Exact achieves MCV, Zaffran et al. (2023a))

If: i) the data is exchangeable, ii) $M \perp\!\!\!\perp X$, iii) $(Y \perp\!\!\!\perp M)|X$, then for almost all imputation function CP-MDA-Exact is such that for any $m \in \{0, 1\}^d$:

$$\mathbb{P}\left(Y \in \hat{C}_\alpha(X, m) | M = m\right) \geq 1 - \alpha,$$

and if additionally the scores are almost surely distinct:

$$\mathbb{P}\left(Y \in \hat{C}_\alpha(X, m) | M = m\right) \leq 1 - \alpha + \frac{1}{\#\text{Cal}^m + 1}.$$

What if we kept all observations?

Test point

3	NA	NA	1
---	----	----	---

Initial calibration set

$x^{(1)}$	-1	-10	6	1
$x^{(2)}$	4	NA	-2	2
$x^{(3)}$	5	1	1	NA
$x^{(4)}$	0	NA	NA	1

Calibration set used

$\tilde{x}^{(1)}$	-1	NA	NA	1
$\tilde{x}^{(2)}$	4	NA	NA	2
$\tilde{x}^{(3)}$	5	NA	NA	NA
$\tilde{x}^{(4)}$	0	NA	NA	1

Introduction to missing values

Quantifying predictive uncertainty with missing values

Conformalized Quantile Regression

Impute-then-Regress+Conformalization

Missing Data Augmentation

MDA with Exact masking

MDA with Nested masking

Experimental results

Conclusion

Idea: modify the test point accordingly

Test point

3	NA	NA	1
---	----	----	---

Initial calibration set

$x^{(1)}$	-1	-10	6	1
$x^{(2)}$	4	NA	-2	2
$x^{(3)}$	5	1	1	NA
$x^{(4)}$	0	NA	NA	1

Calibration set used

$\tilde{x}^{(1)}$	-1	NA	NA	1
$\tilde{x}^{(2)}$	4	NA	NA	2
$\tilde{x}^{(3)}$	5	NA	NA	NA
$\tilde{x}^{(4)}$	0	NA	NA	1

Temporary test points

3	NA	NA	1
3	NA	NA	1
3	NA	NA	NA
3	NA	NA	1

and

$\rightsquigarrow$ similar motivation than Barber et al. (2021)⁷ and Gupta et al. (2022)⁸.

⁷Predictive inference with the jackknife+, The Annals of Statistics

⁸Nested conformal prediction and quantile out-of-bag ensemble methods, Pattern Recognition

5. For a test point $(X^{(n+1)}, M^{(n+1)})$:

5.1 Set $\tilde{M}^{(k)} = \max(M^{(k)}, M^{(n+1)})$ for k
in the **calibration set**

5.2 Impute the new **calibration set**

5.3 For each augmented **calibration point** k :

5.3.1 Get its score $S^{(k)}$

5.3.2 **Impute-then-predict** on the **augmented test point**
 $(X^{(n+1)}, \tilde{M}^{(k)})$, giving: $\widehat{QR}_{\alpha/2}(\tilde{X}^{(n+1),k})$ and
 $\widehat{QR}_{1-\alpha/2}(\tilde{X}^{(n+1),k})$

5.3.3 Compute the corrected prediction interval:

$$[\widehat{QR}_{\alpha/2}(\tilde{X}^{(n+1),k}) - S^{(k)}; \widehat{QR}_{1-\alpha/2}(\tilde{X}^{(n+1),k}) + S^{(k)}] := [Z_{\text{lower}}^{(k)}; Z_{\text{upper}}^{(k)}]$$

5.4 Compute the quantiles $q_{\alpha}(\{Z_{\text{lower}}^{(k)}\}_{k \in \text{Cal}})$ and $q_{1-\alpha}(\{Z_{\text{upper}}^{(k)}\}_{k \in \text{Cal}})$

5.5 Predict $[q_{\alpha}(\{Z_{\text{lower}}^{(k)}\}_{k \in \text{Cal}}); q_{1-\alpha}(\{Z_{\text{upper}}^{(k)}\}_{k \in \text{Cal}})]$

3	NA	NA	1
---	----	----	---

$\tilde{x}^{(1)}$	-1	NA	NA	1
$\tilde{x}^{(2)}$	4	NA	NA	2
$\tilde{x}^{(3)}$	5	NA	NA	NA
$\tilde{x}^{(4)}$	0	NA	NA	1

3	NA	NA	1
3	NA	NA	1
3	NA	NA	NA
3	NA	NA	1

MDA-Nested is Marginally Valid (MV)

Theorem (CP-MDA-Nested marginal validity, Zaffran et al. (2023b))

If the data is exchangeable, then for almost all imputation function CP-MDA-Nested is such that:

$$\mathbb{P} \left(Y \in \hat{C}_\alpha(X, M) \right) \geq 1 - 2\alpha.$$

- ✓ Any missing mechanism (no need to assume $M \perp\!\!\!\perp X$)
- ✓ Does not require $(Y \perp\!\!\!\perp M) | X$
- ✗ Marginal guarantee

Proof element: based on Jackknife+ ideas (Barber et al., 2021).

Leaving-out the k -th data point to predict on the l -th data point

$\leftrightarrow$

Apply the mask of the k -th data point to the l -th data point on which you predict

MDA-Nested (nearly) achieves Mask-Conditional-Validity (MCV)

Stochastic domination of the quantiles (SDQ)

Let $(\check{m}, \check{m}) \in (\{0, 1\}^d)^2$. If $\check{m} \subset \check{m}$ then for any $\delta \in [0, 0.5]$:

$$q_{1-\delta/2}^{Y|(X_{\text{obs}(\check{m})}, M=\check{m})} \leq q_{1-\delta/2}^{Y|(X_{\text{obs}(\check{m})}, M=\check{m})}, \text{ and } q_{\delta/2}^{Y|(X_{\text{obs}(\check{m})}, M=\check{m})} \geq q_{\delta/2}^{Y|(X_{\text{obs}(\check{m})}, M=\check{m})}.$$

$\leadsto$ predictive uncertainty increases with bigger masks.

Theorem (CP-MDA-Nested (nearly) achieves MCV, Zaffran et al. (2023a))

If i) the data is exchangeable, ii) $M \perp\!\!\!\perp X$, iii) $(Y \perp\!\!\!\perp M)|X$, iv) SDQ holds, then for almost all imputation function “CP-MDA-Nested” is s.t. for any $m \in \{0, 1\}^d$:

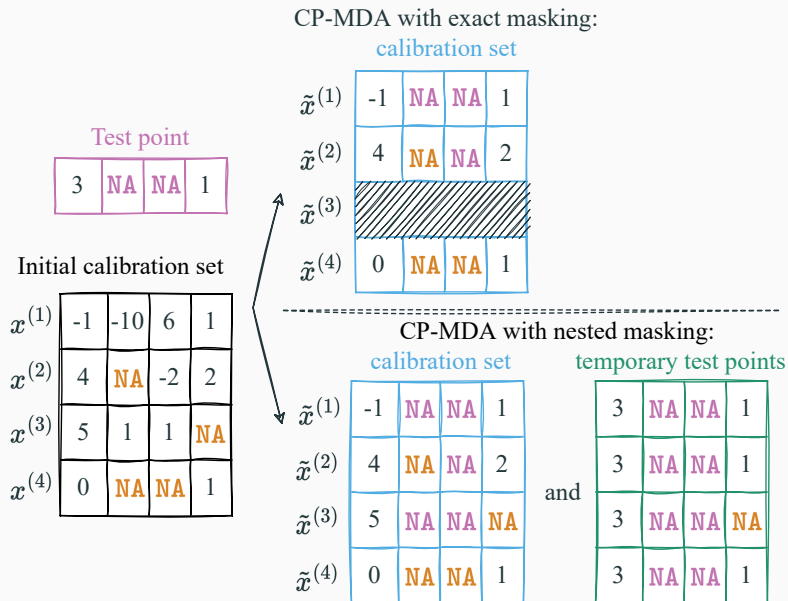
$$\mathbb{P}\left(Y \in \hat{C}_\alpha(X, m) | M = m\right) \geq 1 - \alpha.$$

Change on MDA-Nested: outputs any

$[q_\alpha(\{Z_{\text{lower}}^{(k)}\}_{k \in \text{Cal}^{\check{m}}}); q_{1-\alpha}(\{Z_{\text{upper}}^{(k)}\}_{k \in \text{Cal}^{\check{m}}})]$, where $\check{m}$ is randomly⁹ selected such that $m \subset \check{m}$.

⁹The randomness may depend on $\#\text{Cal}^{\check{m}}$.

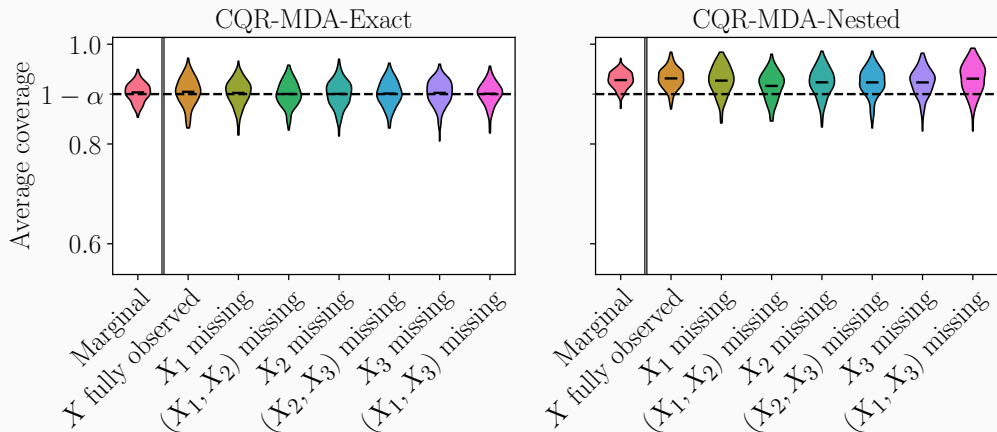
Summary of CP-MDA



MDA achieves Mask-Conditional-Validity (MCV)

$$Y = \beta^T X + \varepsilon,$$

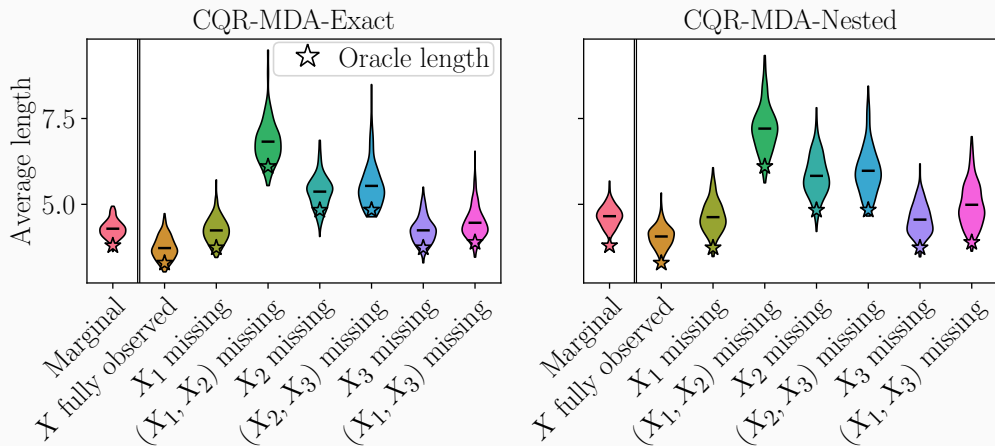
$\beta = (1, 2, -1)^T$, $\varepsilon \perp X$, X and ε Gaussian, 20% uniform MCAR missing values.



MDA achieves (MCV) in an informative way

$$Y = \beta^T X + \varepsilon,$$

$\beta = (1, 2, -1)^T$, $\varepsilon \perp X$, X and ε Gaussian, 20% uniform MCAR missing values.



Introduction to missing values

Quantifying predictive uncertainty with missing values

Conformalized Quantile Regression

Impute-then-Regress+Conformalization

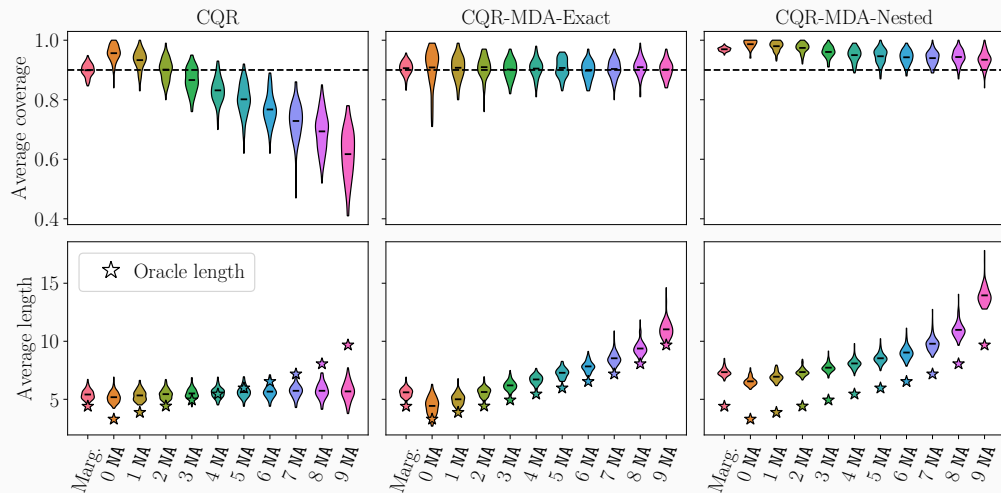
Missing Data Augmentation

Experimental results

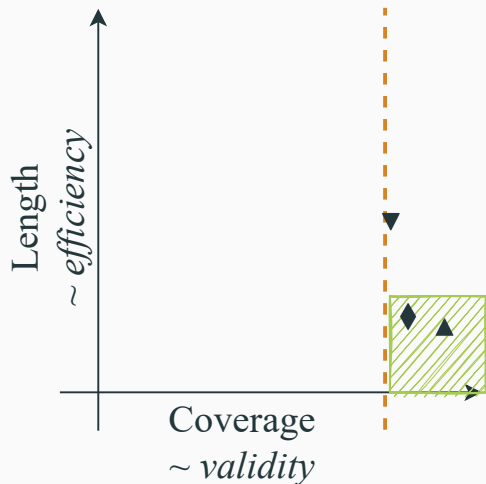
Conclusion

- Imputation by iterative ridge ($\sim$ conditional expectation)
- **Concatenate the mask in the features**
- Neural network, fitted to minimize the pinball loss
- (Semi)-synthetic experiments:
 - Uniform MCAR missing values, with probability 20%
 - 100 repetitions

Synthetic experiments (Gaussian linear model, $d = 10$)



Before more experiments, visualisation



◆ : marginal coverage, i.e.

$$\mathbb{P}(Y \in \hat{C}_\alpha(X, M))$$

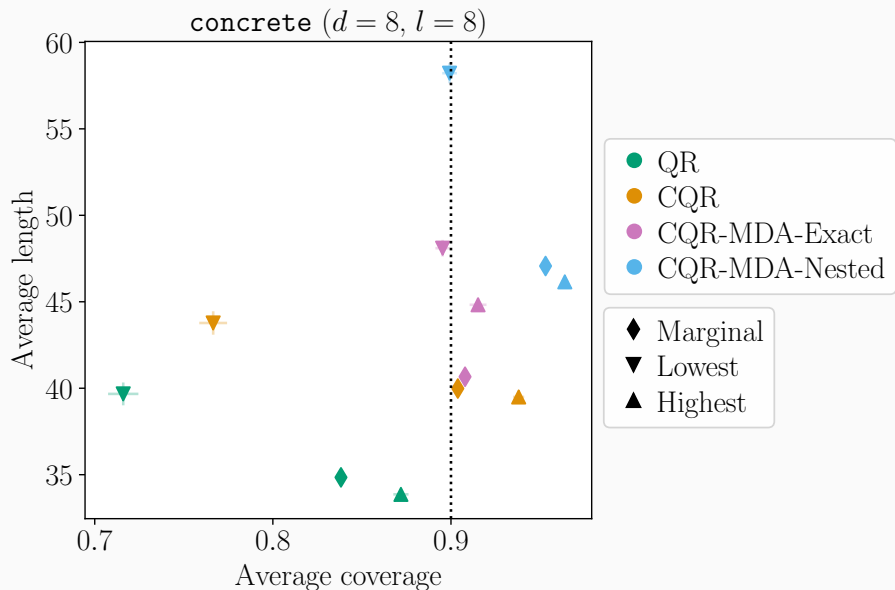
▼ : lowest coverage, i.e.

$$\min_{m \in \mathcal{M}} \mathbb{P}(Y \in \hat{C}_\alpha(X, m) | M = m)$$

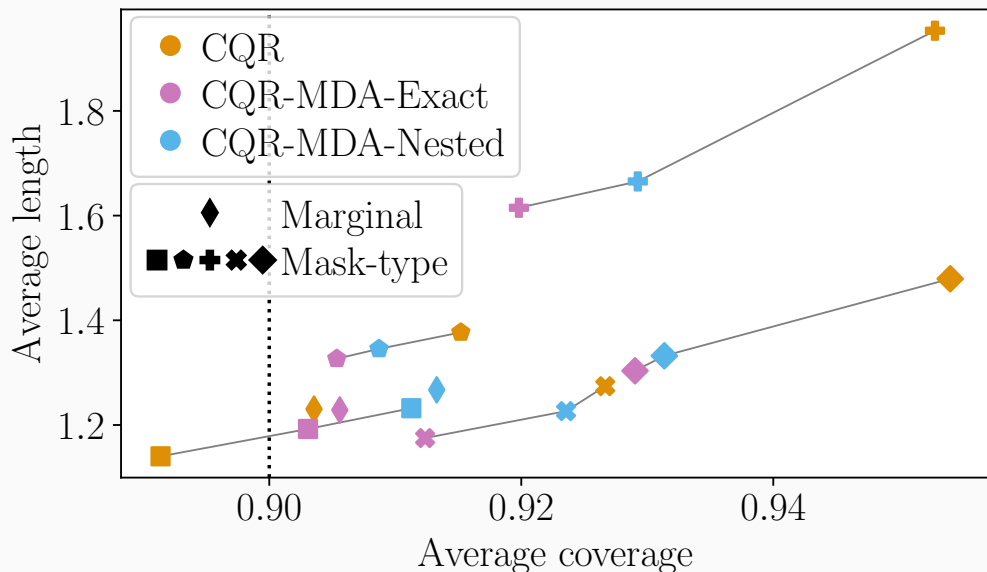
▲ : highest coverage, i.e.

$$\max_{m \in \mathcal{M}} \mathbb{P}(Y \in \hat{C}_\alpha(X, m) | M = m)$$

Semi-synthetic experiments



Real data experiment: TraumaBase[®], critical care medicine



Introduction to missing values

Quantifying predictive uncertainty with missing values

Conclusion

- CP marginal guarantees hold on the imputed data set.
- Missingness introduces additional heteroskedasticity, creating a need for quantile regression based methods.
- CQR fails to attain coverage conditional on the missing pattern.
- **Missing data augmentation is the first method to output predictive intervals with missing values.**
- Missing data augmentation attains conditional coverage with respect to the missing pattern (in MCAR setting).
- Extension: consistency of universal quantile learner when chained with almost any imputation function.



- Investigate alternative methods relying on trade-offs between MDA-Exact and MDA-Nested
- Relationship with Gibbs et al. (2023)¹⁰
 - ✓ Beyond MCAR
 - ✗ Upper bound in $\frac{2^d}{(n+1)\mathbb{P}_M(m)}$: high value for less probable masks
 - ↪ MCV are non-overlapping groups: boils down to splitting the calibration set!
- Quantify the impact of the imputation's choice on Quantile Regression quality in finite sample

¹⁰ *Conformal Prediction With Conditional Guarantees*

Thank you! Questions? :)

- Ayme, A., Boyer, C., Dieuleveut, A., and Scornet, E. (2022). Near-optimal rate of consistency for linear models with missing values. *ICML*.
- Barber, R. F., Candès, E. J., Ramdas, A., and Tibshirani, R. J. (2021). Predictive inference with the jackknife+. *The Annals of Statistics*, 49(1).
- Gibbs, I., Cherian, J. J., and Candès, E. J. (2023). Conformal prediction with conditional guarantees.
- Gupta, C., Kuchibhotla, A. K., and Ramdas, A. (2022). Nested conformal prediction and quantile out-of-bag ensemble methods. *Pattern Recognition*, 127.
- Le Morvan, M., Josse, J., Scornet, E., and Varoquaux, G. (2021). What's a good imputation to predict with missing values? *NeurIPS*.
- Romano, Y., Patterson, E., and Candès, E. (2019). Conformalized Quantile Regression. *NeurIPS*.
- Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63(3).

- Zaffran, M., Dieuleveut, A., Josse, J., and Romano, Y. (2023a). Conformal prediction with missing values. *ICML*.
- Zaffran, M., Dieuleveut, A., Josse, J., and Romano, Y. (2023b). Predictive uncertainty quantification with missing values. To be submitted.
- Zhu, Z., Wang, T., and Samworth, R. J. (2019). High-dimensional principal component analysis with heterogeneous missingness. *arXiv*.

Appendix

Towards asymptotic individualized coverage

Consistency of a universal quantile learner after imputation

Let Φ be an imputation function chosen by the user.

Denote $g_{\beta, \Phi}^* \in \operatorname{argmin}_{g: \mathbb{R}^d \rightarrow \mathbb{R}} \mathbb{E} [\rho_{\beta}(Y - g \circ \Phi(X, M))] := \mathcal{R}_{\beta, \Phi}(g)$.

Comparison with: $\operatorname{argmin}_f \mathbb{E} [\rho_{\beta}(Y - f(X, M))] \text{ (informal)}$.

Proposition (Pinball-consistency of an universal learner)

For almost all $\mathcal{C}^{\infty}$ imputation function Φ , the function $g_{\beta, \Phi}^* \circ \Phi$ is Bayes optimal for the pinball-risk of level β .

$\hookrightarrow$ any universally consistent algorithm for quantile regression trained on the data imputed by Φ is pinball-Bayes-consistent.

This is an extension of the result of Le Morvan et al. (2021).

Corollary

For any missing mechanism, for almost all C^∞ imputation function Φ , if $F_{Y|(X_{\text{obs}(M)}, M)}$ is continuous, a universally consistent quantile regressor trained on the imputed data set yields asymptotic conditional coverage.

$\hookrightarrow \mathbb{P}(Y \in \hat{C}_\alpha(x) | X = x, M = m) \geq 1 - \alpha$ for any $m \in \mathcal{M}$ and any $x \in \mathbb{R}^d$, asymptotically with a super quantile learner.

$$d = 3$$

Data generation

$$(X, Y) \in \mathbb{R}^3 \times \mathbb{R}.$$

$$Y = \beta^T X + \varepsilon$$

with $\varepsilon \sim \mathcal{N}(0, 1)$, $\beta = (1, 2, -1)^T$ and

$$(X_1, X_2, X_3) \sim \mathcal{N} \left(\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 & 0.8 & 0.8 \\ 0.8 & 1 & 0.8 \\ 0.8 & 0.8 & 1 \end{pmatrix} \right).$$

All components of X each have a probability 0.2 of being missing, Completely At Random.

Simulation settings

- Method: CQR
- Basemodel: neural network
- 200 repetitions
 - train size of 250 points
 - calibration size of 250 points
 - test size of 2000 points

$d = 10$, with missing data augmentation

Data generation

$$(X, Y) \in \mathbb{R}^{10} \times \mathbb{R}.$$

$$Y = \beta^T X + \varepsilon$$

with $\varepsilon \sim \mathcal{N}(0, 1)$, $\beta = (1, 2, -1, 3, -0.5, -1, 0.3, 1.7, 0.4, -0.3)^T$ and

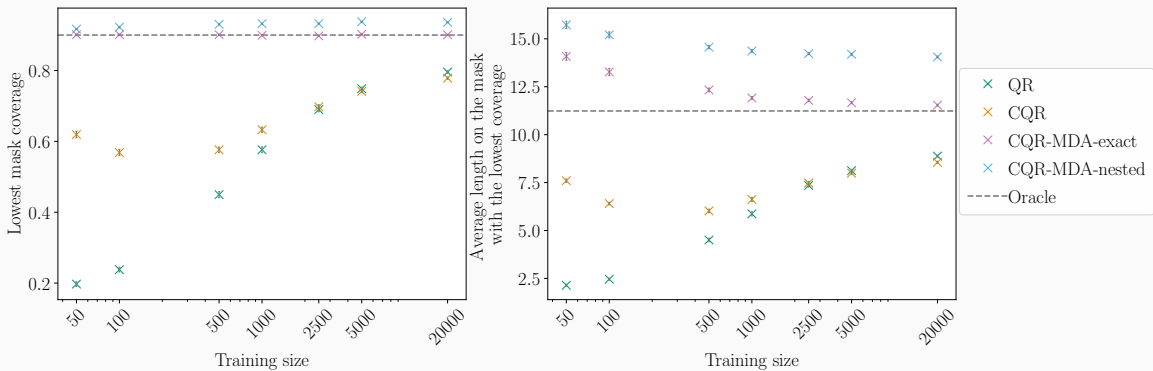
$$(X_1, \dots, X_{10}) \sim \mathcal{N} \left(\begin{pmatrix} 1 \\ \vdots \\ \vdots \\ 1 \end{pmatrix}, \begin{pmatrix} 1 & 0.8 & \cdots & 0.8 \\ 0.8 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0.8 \\ 0.8 & \cdots & 0.8 & 1 \end{pmatrix} \right).$$

All components of X each have a probability 0.2 of being missing, Completely At Random.

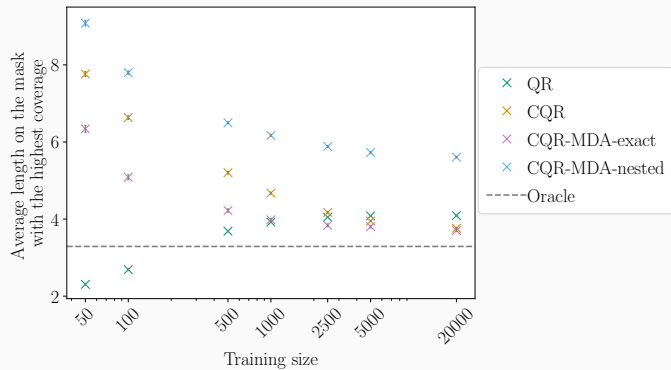
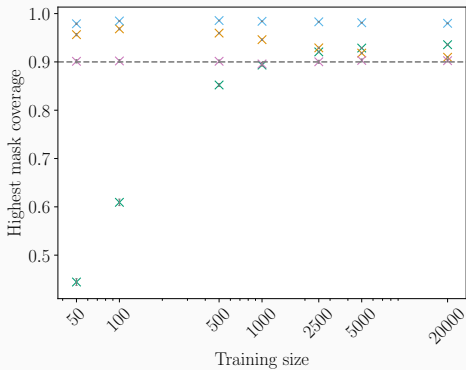
Simulation settings: varying training size

- Method: CQR
- Basemodel: neural network
- Imputation: iterative ($\approx$ conditional expectation)
- Mask as features: yes
- 100 repetitions
 - train size varies
 - calibration size of 1000 points
 - test size of 2000 points

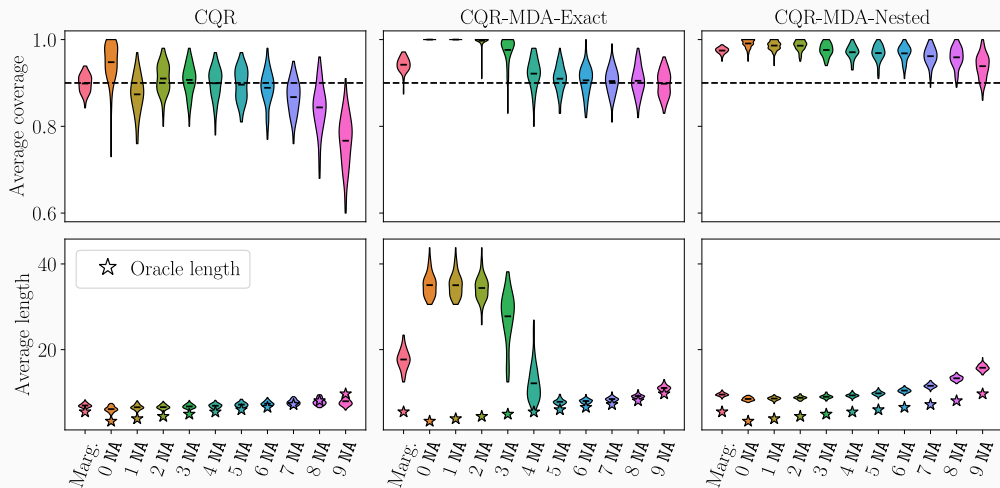
Results on the worst group



Results on the best group



Synthetic experiments, 40% of missing values (Gaussian linear model, $d = 10$)

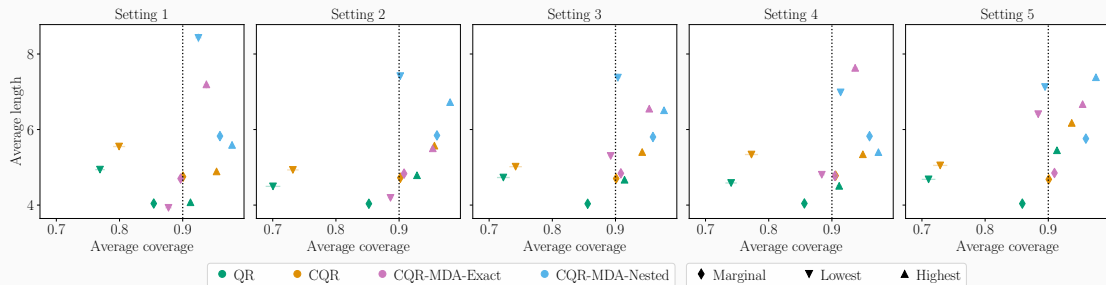


Simulation settings: beyond MCAR

- 6 variables (denote this set X_{missing}) out of 10 can be missing (the 4 others form the set X_{observed})
→ $X_{\text{missing}} = \{X_1, X_2, X_3, X_5, X_8, X_9\};$
- Proportion of missing entries fixed to be 20%.

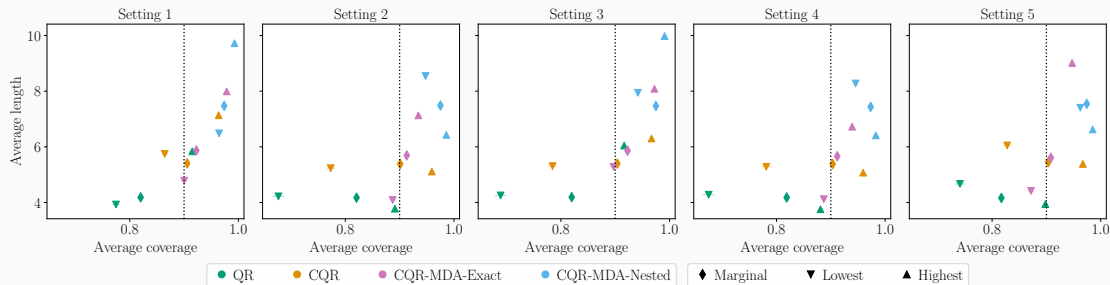
MAR missingness

- Probability of the variables in X_{missing} to be missing given by a logistic model of arguments X_{observed} .
- This setting is declined 5 times, with different weights for the logistic model.



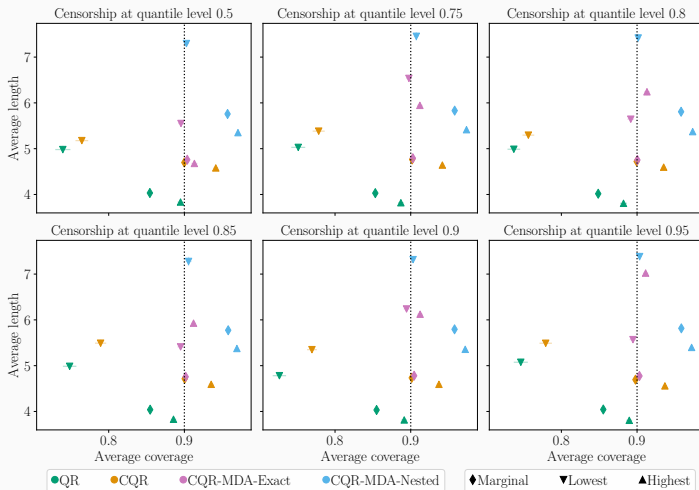
MNAR self masked missingness

- Probability of each variable in X_{missing} to be missing given by a logistic model of argument the same variable of X_{missing} .
- This setting is declined 5 times, with different weights for the logistic model.

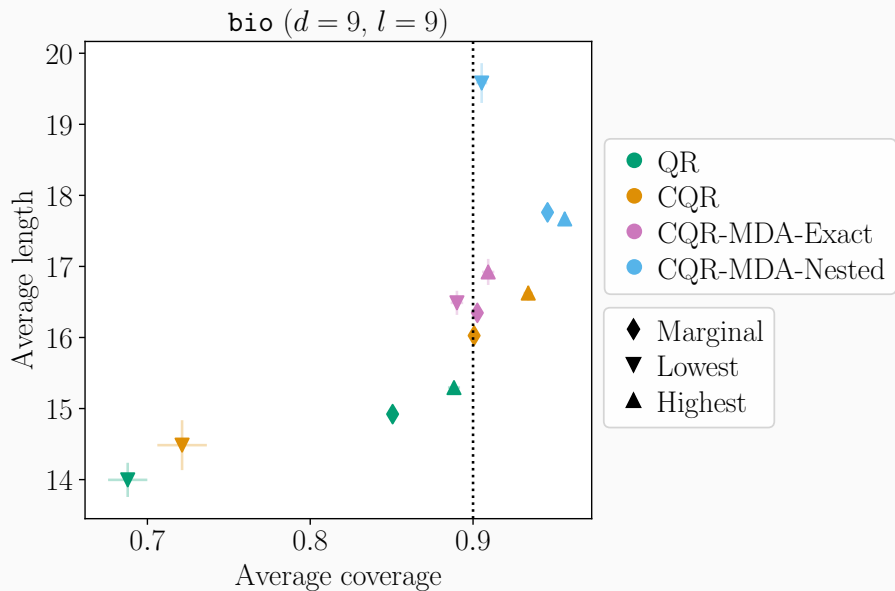


MNAR quantile censorship missingness

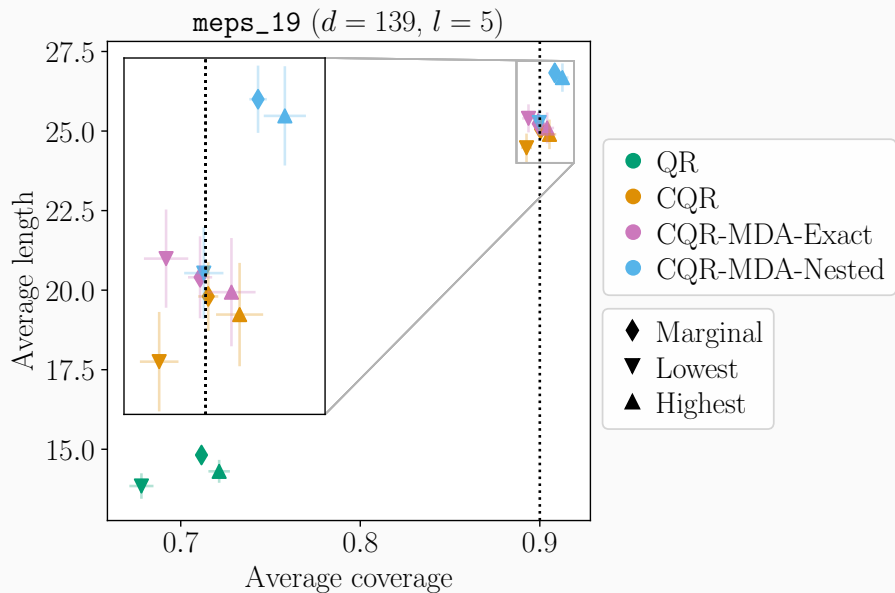
- Missing values are introduced at random in each q -quantile of the variables in X_{missing} .
- 6 different settings: q varies between 0.5, 0.75, 0.8, 0.85, 0.9 and 0.95.



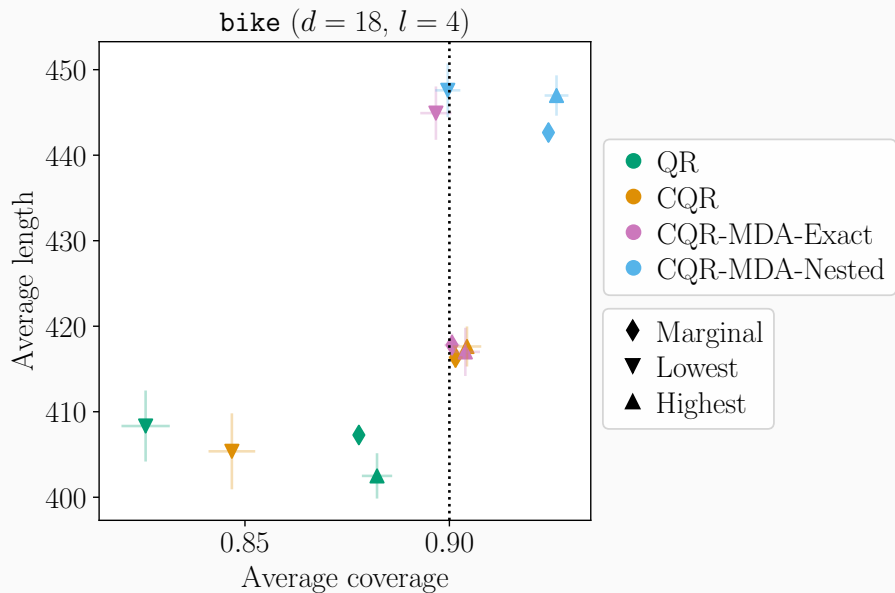
Semi-synthetic experiments



Meps_19 data set



Bike data set



TraumaBase®

Data set description i

- Age: the age of the patient (no missing values);
- Lactate: the conjugate base of lactic acid, upon arrival at the hospital (17.66% missing values);
- Delta_hemo: the difference between the hemoglobin upon arrival at hospital and the one in the ambulance (23.82% missing values);
- VE: binary variable indicating if a Volume Expander was applied in the ambulance. A volume expander is a type of intravenous therapy that has the function of providing volume for the circulatory system (2.46% missing values);
- RBC: a binary index which indicates whether the transfusion of Red Blood Cells Concentrates is performed (0.37% missing values);

- SI: the shock index. It indicates the level of occult shock based on heart rate (HR) and systolic blood pressure (SBP), that is $SI = \frac{HR}{SBP}$, upon arrival at hospital (2.09% missing values);
- HR: the heart rate measured upon arrival of hospital (1.62% missing values).